

5(a)	<table><tr><th>feature</th><th>ADH</th><th>glucagon</th><th>insulin</th><th></th></tr><tr><td>binds to receptors on cell surface membranes</td><td>✓</td><td>✓</td><td>✓</td><td>;</td></tr><tr><td>results in molecules moving from cells into the blood</td><td>✓</td><td>✓</td><td>x</td><td>;</td></tr><tr><td>is secreted as a result of detection by osmoreceptors</td><td>✓</td><td>x</td><td>x</td><td>;</td></tr></table>	feature	ADH	glucagon	insulin		binds to receptors on cell surface membranes	✓	✓	✓	;	results in molecules moving from cells into the blood	✓	✓	x	;	is secreted as a result of detection by osmoreceptors	✓	x	x	;	3
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5(b)	<i>any four from:</i> 1 impulses / action potentials / neurotransmitters, in nervous system <u>and</u> hormones in endocrine system ; 2 electrical in nervous system <u>and</u> chemical in endocrine system ; 3 neurones / nerve cells, in nervous system <u>and</u> blood in endocrine system ; 4 effects are / target is, specific / localised, in nervous system <u>and</u> (can be) widespread in endocrine system ; 5 effect / response, happens quickly in nervous system and slowly in endocrine system ; 6 effect / response, lasts a short time in nervous system <u>and</u> long time in endocrine system ; 7 AVP ;	4																				
5(c)(i)	C ;	1																				
5(c)(ii)	B ;	1																				

Question	Answer	Marks
5(c)(iii)	1 (D / it), shortens / contracts / decreases ; <i>and any three from:</i> 2 calcium ions / Ca^{2+}, bind to troponin ; 3 tropomyosin, moves / shifts position / exposes binding sites on actin ; 4 myosin (head) binds to actin / myosin-actin cross-bridges form ; 5 power stroke / myosin head, pulls actin ; 6 more actin, enters / overlaps with, A band / B or more overlap between actin and myosin ;	4

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